



Meeting Water Supply Challenges in the Colorado River Basin

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Executive Summary

Water resource management failures in the Colorado River Basin risk long-term economic decline as water use rapidly eclipses available supply. Sustainability of the Basin's water resources is threatened as farmers receive subsidies for revenue losses in drought years, Basin states and the federal government struggle over apportionment authority, and prior appropriation mandates full use of available water whether or not need exists. Congress and Basin resource officials must intervene to modernize management schemes and ensure sustainability of water resources for future generations. The recommended course of action is to advance ongoing drought contingency efforts while also pursuing a three-pronged strategy that updates federal Farm Bill subsidy structures, establishes an independent regional water banking and management authority, and abolishes the forfeiture aspect of prior appropriation.

Keywords: *Colorado River, Water Resource Management, Water in the West, Reclamation, Drought, Water Bank*

The Issue: Mismanagement and Resource Imbalances

Water resource imbalances in the Colorado River Basin risk economic decline across the American Southwest. An excess of demand over decreasing water supply, these resource imbalances threaten the long-term sustainability of megalopolises and rural areas in the Colorado River Basin. The Basin - which includes the entire state of Arizona, Southern California, and parts of Nevada, New Mexico, Colorado, Utah, and Wyoming - benefits from the agricultural, municipal, and hydropower capabilities of the Colorado River. However, decades of mismanagement and ongoing policy failures have strained the Colorado River's balance of water supply and demand beyond sustainable levels (Clowes, Hustead, & Kolomitz, 2016). The resource imbalance is so advanced, the lower 100 miles of the Colorado River has dried to the

point that it no longer reaches the sea except in years of heavy runoff (James et al., 2014). The effect of mismanagement on the water resource imbalance in the Colorado River Basin is coupled by rapid population growth and an ongoing drought, which ranks as the worst in recorded history (Buschatzke & Klobas, 2018).

Given current management strategies in the Basin, cities like Las Vegas, Phoenix, Denver, and Los Angeles face constant or slightly decreasing volumes of water supply and rapidly increasing water demand over the next 50 years (USBR, 2012; See *Figure 1*). A 2012 United States Bureau of Reclamation forecast uses a 10-year running average calculation to approximate water supply and demand levels until the mid-21st century (USBR, 2012). *Figure 1* shows these metrics at five-year intervals for the years 1923-2063. The 10-year running average approach averages the preceding 10 years of data for each consecutive point and plots that value in the corresponding location. In other words, each point reflects the average of the preceding 10 years of data. The dotted line at 2008 signals the demarcation between historical data and projected values. Per the study, demand exceeded supply for the first time in the early 21st century and will continue to grow in magnitude into the 22nd century and beyond (USBR, 2012; See *Figure 1*).

Figure 1: Water Supply and Demand in the Colorado River Basin

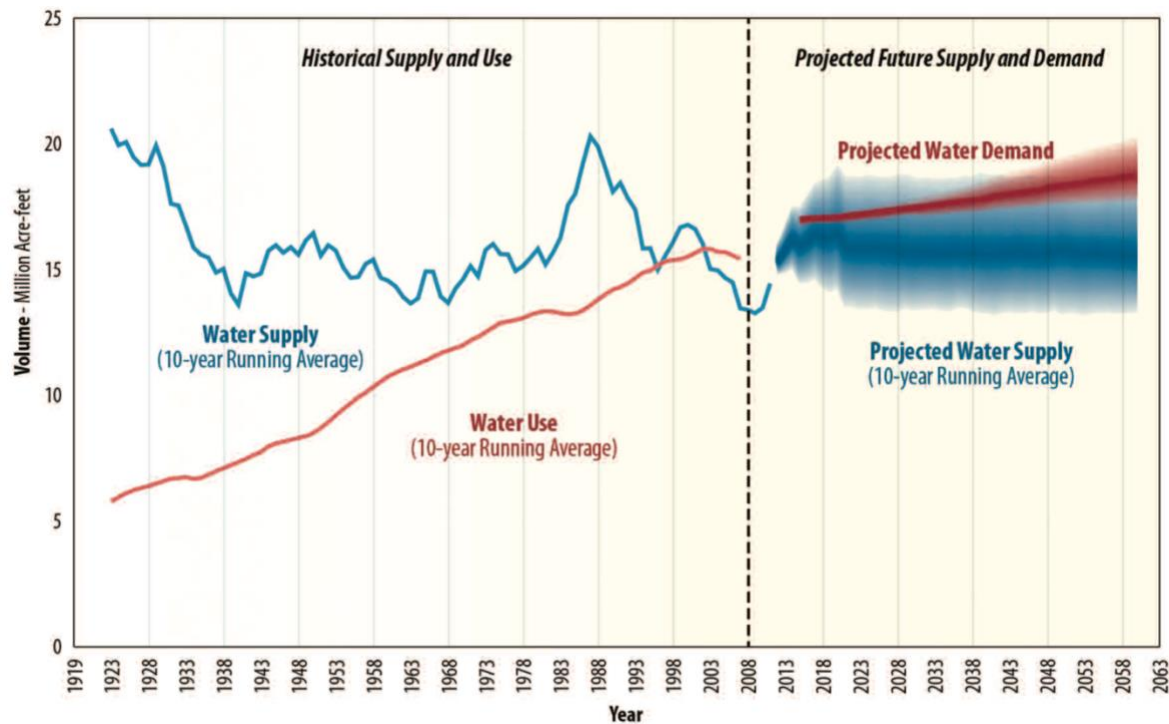


Figure 1, United States Bureau of Reclamation, 2012

Rationale for Action

The Colorado River generates an estimated \$1.43 trillion in GDP annually (James et al., 2014). This accounts for nearly 65% of the Basin Region's total GDP and one twelfth of the GDP for the entire United States (James et al., 2014). In addition, the river accounts for over 16 million jobs and \$871.45 billion in labor incomes per year across the seven Basin states (James et al., 2014). Due to the region's nearly singular dependence on the Colorado River (and the Basin's accompanying groundwater) as a source of water, a 10% reduction of available water in the Colorado River would result in an estimated \$143 billion in losses across the Basin (James et al., 2014). Basin residents could feel the short-run effects of these shocks through sustained power outages and millions of lost jobs. In the long run, Colorado River Basin economies and many others throughout the United States, who depend on the Basin for the majority of

agriculture produced in the U.S., could see resource price increases and even emergency resource rationing (Clowes, Hustead, & Kolomitz, 2016).

Economic Importance of the Colorado River by Sector

Agriculture

Despite the arid climate in the Southwest, the Colorado River Basin is one of the most productive agricultural regions in the world (Cooley et al., 2016). Basin states produce nearly 20% of all agricultural commodities and livestock in the United States, generating over \$72 billion in agricultural revenues per year (Cooley et al., 2016; “Cash Receipts,” 2018). Given the lack of precipitation in the region, Basin farms draw the majority of their water from the Colorado River (Clowes, Hustead, & Kolomitz, 2016; Cooley et al., 2016). The Colorado River irrigates over 5.5 million acres of farmland across the Basin states and agricultural withdrawals from the Colorado River account for an estimated 70% of agricultural withdrawals from all rivers in the United States (USBR, 2012; Cooley et al., 2016).

Municipal

The Colorado River provides water for municipal use to over 40 million people in the Colorado River Basin (USBR, 2012). This includes the water supply for five of the top ten most populous counties in the country: Los Angeles County (1st, CA), Maricopa County (4th, AZ), San Diego County (5th, CA), Orange County (6th, CA), and Riverside County (10th, CA) (Census Bureau, 2018). These counties combined for a GDP of more than \$1.6 trillion in 2015 (Panek & Baumgardner, 2018). Additionally, the Southwest is the country’s fastest-growing region; Maricopa County, Clark County (NV), and Riverside County are the top three fastest-growing counties in the United States (Cooley et al., 2016; Census Bureau, 2018). The rapidly growing population of the Colorado River Basin is set to impose additional pressure on the region’s water resources.

Hydropower

The large flow and steep gradient of the Colorado River make it ideal for generating hydroelectric power (James et al., 2014). The river generates an estimated 4,200 megawatts of electrical power throughout the Colorado River Basin (USBR, 2012). Given the various intricacies of the Basin river network, there exist both market and non-market values and externalities for hydropower in the Southwest (Lowry et al., 2017). These include the electrical power generated by the river and revenues that result from use of that electricity, as well as negative, non-market externalities such as morbidity or aesthetic effects from power plants along the river (Lowry et al., 2017). From an economic frame of reference, the Colorado River's use for hydropower includes opportunity costs and Pareto inefficiency (Lowry et al., 2017). For example, optimal changes in dam operations at one point in the river may cause significant changes in water or power prices for rural communities at another point in the river (Lowry et al., 2017). However, the river's use for hydroelectric power generation is beneficial in that it offsets fossil fuel contributions in the Southwest and provides renewable power to millions of people (USBR, 2012).

Causes of Water Resource Imbalances in the Southwest

While the effects of the drought and population growth on water resources in the Colorado River Basin are well-documented and widely accepted (USBR, 2012; Yigzaw & Hossain, 2016; Buschatzke & Klobas, 2018), the primary purpose of this brief is to provide recommendations regarding regional water management schemes as well as federal and regional water policies. To this end, the causes of water resource imbalances in the Basin discussed encompass failures in the aforementioned management schemes and policies enacted to date. These are as follows:

Farm Bill and Agricultural Policy

United States water policy contributes to the water resource imbalance in the Southwest by prioritizing agriculture subsidies over water conservation. The United States Farm Bill is an omnibus law that has governed a variety of food and agricultural issues since the 1930's (Johnson & Monke, 2017). The law is updated and reauthorized every few years, most recently with the Agricultural Improvement Act of 2018 (Stein, 2018). While the Farm Bill has supported the food and agriculture industries and administered federal food stamp programs effectively for decades, the law also includes management strategies that have been detrimental to water conservation in the Southwest (Clowes, Hustead, & Kolomitz, 2016).

These strategies include the provision of agriculture subsidies to farmers in case of revenue losses from drought (Clowes, Hustead, & Kolomitz, 2016). While the government collects a premium on Farm Bill drought insurance, farmers in the Southwest received \$1.90 for every \$1 of premium in 2011 due to the ongoing drought (Clowes, Hustead, & Kolomitz, 2016). By insuring farmers against drought conditions, the federal Farm Bill disincentivizes water conservation; Farmers can continue to use the maximum allotment of water provided to them under water licensing agreements with no regard for natural conditions (Clowes, Hustead, & Kolomitz, 2016). In fact, some of the most water-intensive crops are grown in the Colorado River Basin; California produces 99% of the nation's walnuts and 95% of its broccoli (Park & Lurie, 2014).

Federalism and State Policy Conflicts

Colorado River Basin states and the United States government have long struggled with federalism as a means for determining authority over water apportionments in the Colorado River and its accompanying tributaries (Griggs, 2017). The 1922 Colorado River Compact

established a framework for equitable distribution of water rights among states and tribes in the Colorado River Basin (Chapman & Denver, 2013; Griggs, 2017). The agreement split the seven Basin states into two sub-basins, allocating water rights totaling 7.5 million acre-feet annually to each division. The Compact left apportionments within each sub-basin to the states themselves. However, Arizona's concern that California would monopolize rights to the sub-basin apportionment for the Lower Basin led Arizona to refuse to ratify the Colorado River Compact.

The struggle for water rights and apportionment authority in the Colorado River Basin was further entrenched through the 1928 Boulder Canyon Project Act and subsequent lawsuits and agreements (Griggs, 2017). The Boulder Canyon Project Act created a federally mandated apportionment scheme for the entire Colorado River, redistributing apportionments outlined in the Colorado River Compact to the states rather than the sub-basins (H.R. 5773, 1928). In 1963, the Supreme Court's landmark *Arizona v. California* decision awarded states the right to appropriate and use tributary flow within state borders before the tributary comingled with the Colorado (Chapman & Denver, 2013). This ruling settled a decades-long dispute between Arizona and California, holding that the Boulder Canyon Project Act applied only to the main channel of the Colorado River and that tributary apportionments within state borders were the responsibility of the underlying state (Chapman & Denver, 2013).

The 1922 Colorado River Compact, Boulder Canyon Project Act of 1928, and the *Arizona v. California* case are only a few examples of the struggle for apportionment authority in the Colorado River Basin. The failure of the Basin states and the federal government to determine an effective co-management strategy contributes to water resource imbalances in the Basin as resources are dealt by inefficient, zero-sum management schemes. Further, resource

managers are constantly at odds with one another and operations are encumbered by frequent litigation.

Prior Appropriation

The doctrine of prior appropriation has guided the “Law of the River” (the cumulative body of interstate compacts, federal and state statutes, and court decisions regarding Colorado River apportionment) for over 150 years (Hess, 1917). The doctrine was established by a series of California laws enacted in the early 1850’s and it was upheld via the 1855 California Supreme Court case, *Irwin v. Phillips* (Hess, 1917). Prior appropriation awards water rights to users who put water to a “beneficial and productive use” (Clowes, Hustead, & Kolomitz, 2016). These rights are subject to forfeiture should the owner fail to use the allotted water for a statutory period (Clowes, Hustead, & Kolomitz, 2016). Prior appropriation differs from riparian rights, used in the rest of the United States, in that riparian rights grant water rights to land owners along a river; prior appropriation prioritizes intended uses of water in determining rights (Hess, 1917; Clowes, Hustead, & Kolomitz, 2016; Griggs, 2017).

While prior appropriation has clear benefits (incentivizing irrigation and hydropower generation), the doctrine actually disincentivizes water conservation (Clowes, Hustead, & Kolomitz, 2016). As water rights granted under prior appropriation are subject to forfeiture if an owner fails to use their full allotment, water users have no incentive to decrease water use. In fact, water users must make full use of their appropriated water, even if they do not need the full allotment, to ensure their rights remain intact. The ‘use it or lose it’ policy advanced under prior appropriation encourages wasteful water practices and disincentivizes water conservation.

Current Approach/Status Quo

Ongoing efforts to address growing water resource imbalances in the Colorado River Basin prioritize drought mitigation above modernization of management strategies and policies regarding apportionment of Colorado River water rights. The United States Bureau of Reclamation and the seven Basin states are working to advance the Drought Contingency Plan (DCP), an interstate compact to incentivize water storage and avoid catastrophic reductions in water supply throughout the Basin (Buschatzke & Klobas, 2018). While the DCP is an important first step in ensuring long-term sustainability of the Colorado River Basin, it fails to address systemic policy and management failures ingrained in the region's appropriation scheme. The DCP will likely reduce the impact of the ongoing drought on water resources in the Colorado River Basin; however, the plan does not implement management strategies or policy amendments that ensure sustained reductions of the projected water resource deficit in the Basin region.

Proposed Policy Solutions

Congress and Basin resource managers must take further action to modernize Colorado River Basin management schemes and ensure water supply meets demand for centuries to come. Each policy solution reviewed accounts for population growth and drought, as well as addresses one or more of the management failures discussed above. The following policy solutions are considered:

1. Update federal Farm Bill subsidy structures
2. Establish an independent regional water banking and management authority
3. Abolish forfeiture policy and incentivize surplus water rights transfers

Farm Bill subsidy structure updates

Congress should update the federal Farm Bill subsidy structure to prioritize water conservation in determining reimbursement of losses. Under the current structure, subsidies fail to create incentives to use more water-efficient agricultural technologies (Clowes, Hustead, & Kolomitz, 2016). These subsidies are set to cost taxpayers \$94 billion between 2013 and 2023 (Nixon, 2013). Though they are expensive and detrimental to water conservation in the Colorado River Basin, subsidies to farmers are extremely politically popular in the legislature (Clowes, Hustead, & Kolomitz, 2016). To both maintain crop insurance and reduce the water resource imbalance in the Southwest, Congress should update the Farm Bill subsidy structure to require investment in water-efficient technologies and implementation of agricultural strategies that prioritize water conservation. Through this management scheme, Congress will ensure long-term sustainability of water resources in agriculture as well as save costs in the long run due to the effects of water conservation.

Independent regional water banking and management authority

Congress, the United States Bureau of Reclamation (USBR), and the Basin states should establish an independent regional water banking and management authority to overcome the negative effects of federalism and state policy conflicts on water resource management in the Colorado River Basin. A water bank is an institutional mechanism used to facilitate short-term transfers of water rights between users (Castle & MacDonnell, 2016). This reduces incentives for wasteful irrigation practices and allows excess water rights to be loaned (for a fee) between users. An independent regional authority would be administered by representatives from the USBR and the seven Basin states, and it would be overseen by the relevant Congressional committee. A regional water banking and management authority will ensure long-term economic security in the face of resource imbalances in the Colorado River Basin.

Surplus water rights transfers

Congress should abolish the ‘use it or lose it’ forfeiture policy advanced under prior appropriation and incentivize surplus water rights transfers. This amendment to prior appropriation will promote water conservation and expand the supply of available water in the Colorado River Basin. The present forfeiture policy disincentivizes water conservation by requiring full use of surplus water, even if such use is not necessary (Clowes, Hustead, & Kolomitz, 2016). While it would be detrimental to rescind the doctrine of prior appropriation as a whole, Congress can advance efforts to reduce the water deficit by removing the forfeiture policy and allowing rights owners to transfer surplus rights without risk of losing their own appropriation. Surplus water rights transfers ensure a more sustainable water supply, as water users would trade, rather than wastefully use, excess resources. These surplus transfers would be entered into the water bank system, further contributing to water deficit reduction.

Recommendation and Conclusion

In addition to maintaining the status quo and advancing the Drought Contingency Plan, Congress and the Basin states should take a three-pronged approach to water resource management in the Colorado River Basin: update federal Farm Bill subsidy structures, establish an independent regional water banking and management authority, and abolish the wasteful forfeiture policy. These policy measures are necessary to reduce demand and modify management strategies with an emphasis on resource sustainability. Given current management failures and negative resource projections, Congress and Basin resource managers must act quickly and decisively to ensure long-term economic and functional sustainability of the Colorado River Basin.

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